Lawrence Du, Ph.D.

Machine Learning • Biology • MLOps
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Skills

Languages
Python, SQL, C/C++, C#, Rust, Java, Dart, Bash, HTML/CSS, MATLAB

ML and data tools
PyTorch, TensorFlow, scikit-learn (sklearn), NumPy, SciPy, pandas, Numba, Ray, MLflow, Flask, Jupyter

Cloud and MLOps
Amazon Web Services (AWS: EC2, S3, Batch, Glue, Athena, Fargate), Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, Pulumi, Metaflow, GitHub Actions, Jenkins, Linux

ML and scientific computing
Low-rank adaptation (LoRA), sparse autoencoders (SAE), Joint Embedding Predictive Architecture (JEPA), foundation model fine-tuning, deep learning, transformers, convolutional neural networks (CNNs), graph neural networks (GNNs), GANs, diffusion, Louvain/Leiden community detection, SVM, PCA, KNN, decision trees, cross-validation

Experience

GSK • Senior AI/ML Engineer, Sequence Learning and Phenomics
Oct 2024 – Present | South San Francisco, CA

- Built an end-to-end fine-tuning platform for sequence foundation models, enabling internal teams to predict RNA properties from nucleotide sequences and perturbation responses from single-cell RNA-seq.
  Cut iteration time **from weeks to less than a day**.
- Integrated Cellpose-SAM, combining Cellpose and the Segment Anything Model (SAM), into production computer vision and image-embedding pipelines for image segmentation, enabling single-cell tracking and cell-state world modeling across **millions of images**.
- Built a zero-shot image-embedding model to predict iPSC-to-microglia differentiation success, scoring **>3 million cells** for active Alzheimer's disease research and saving **up to six weeks of labor** on two-month differentiation protocols.
  Used sparse autoencoders to extract interpretable features.
- Piloted generative AI using x-prediction / JiT diffusion models to generate protein subcellular localization images conditioned on ESM protein embeddings.
- Built a pipeline using large language models (LLMs) for retrieval-augmented generation (RAG) and gene-candidate evaluation.
  Prioritized **~1,000 candidate genes** for a lung-epithelium CRISPR knockout screen supporting a next-generation COPD drug program.
- Shipped Model Context Protocol (MCP) servers that let company-wide LLM assistants build and launch scientists' image-analysis pipelines, making internal AI Science tools agent-callable and accelerating analysis turnaround **>10×**.
- Contributing to benchmarking tasks for open-weight and commercial LLMs (currently in development); led software engineering interviews to support hiring on the team.

Freenome • Senior Machine Learning Research Engineer
Aug 2022 – Jun 2024 | South San Francisco, CA

- Led a greenfield distributed ML platform project using PyTorch, Ray, and Kubernetes for cancer detection from methylated DNA and protein data.
  Enabled larger models and accelerated model training **>10×** using distributed data parallelism (DDP).
- Deployed organization-wide MLflow experiment tracking on GCP using Terraform and Pulumi.
  Enabled live training monitoring, results sharing, and automated reproducible reports, reducing manual researcher effort **≥5×**.
- Built scalable multitask neural networks and elastic net models in PyTorch for colorectal cancer risk from cell-free DNA, supporting a clinical trial cohort of **>27,000 individuals**.

23andMe • Machine Learning Engineer
Apr 2020 – Aug 2022 | Sunnyvale, CA

- Built large-scale data pipelines for quality statistics and feature engineering of genotyped and imputed single-nucleotide polymorphisms (SNPs).
  Implemented extract, transform, load (ETL) workflows for **~10 million samples × ~1 million SNPs** using AWS Batch, Metaflow, and Glue, populating an Athena feature datastore for genome-wide association studies (GWAS) and polygenic risk scores (PRS).
- Improved type 2 diabetes and coronary artery disease PRS models through stacking ensembles in production pipelines, improving sensitivity and specificity for **>400,000 customers**. Evaluated performance using AUC and auPRC.
- Automated model evaluation and classifier reports with MLflow artifact storage and headless Jupyter execution, reducing researcher analysis time **from days to minutes**.

23andMe • Data Scientist, Ancestry Product
Nov 2018 – Apr 2020 | Sunnyvale, CA

- Created and deployed Recent Ancestor Locations (RAL), a high-precision, high-recall country-matching algorithm serving **>15 million customers** worldwide. Deployed the model as an AWS Fargate microservice, using MLflow for experiment tracking and a model repository.
- Improved graph-based techniques for identifying genetic populations from identity-by-descent (IBD) family relationships. Used graph community detection to identify subpopulations in Mexico and the United Kingdom.

Scripps Research • Bioinformatician IV
May 2018 – Oct 2018 | San Diego, CA

- Developed an organ transplant rejection classifier from RNA sequencing data for a research grant.
- Built automated Oxford Nanopore long-read bioinformatics pipelines in Common Workflow Language (CWL) for real-time microbial diagnostics, *de novo* genome assembly, and variant calling.

Juno Diagnostics · Independent Consultant

Sep 2017 – Feb 2018 | San Diego, CA

- Developed a TensorFlow deep learning software package for detecting prenatal genetic abnormalities from high-throughput sequencing data; co-invented US20210020314A1: Deep learning-based prenatal testing.
- Created a genetic-abnormality simulator derived from statistical analysis of human high-throughput sequencing data.

Insight · Data Science Fellow

Jan 2017 – Apr 2017 | Remote / San Diego, CA

- Built DeepPixelMonster, a TensorFlow deep convolutional generative adversarial network (DCGAN) for pixel art, and deployed it as an interactive Flask application on AWS EC2.

UC San Diego · Ph.D. Researcher, Biology · Scott A. Rifkin Lab

Aug 2010 – May 2017 | La Jolla, CA

- Built DeepNuc, a CNN for classifying **>500,000** transcription start site (TSS) flanking sequences from humans, mice, fruit flies, and nematodes, plus **>60,000** microRNA target sequences from publicly available CLEAR-CLIP data.
- Studied developmental RNA expression noise in **>5,000 embryos** using single-molecule imaging and custom MATLAB tools for image segmentation, fluorescence quantification, and deconvolution.
- Generated **>70 transgenic nematode and Drosophila strains** using Gibson assembly, CRISPR/Cas9, MosSCI, and PhiC31 integrase.

Education

Ph.D., Biology · UC San Diego
2010–2017 • Scott A. Rifkin Lab

B.A., Biological Sciences · Cornell University
2006–2010 • Genetics and Development • *magna cum laude*

Side Projects

- **Rogue Stargun** — Unity/C# VR game with **>10,000 installs** across Steam, Quest, and PICO; **4.7/5 on Meta Quest**. Frontier LLMs and ElevenLabs text-to-speech in **nine languages**; dogfighting AI with collision avoidance, firing intercept, and behavior trees for **40+ concurrent ships**. Shader and rendering/batch optimization: **72 FPS on Quest 2, 90+ FPS on Quest 3 and PICO Ultra 4**
- **DiffuMon** — Open-source PyTorch DDPM/DDIM image generator and paper reproduction for commodity GPUs.
- **DuBlog** — Technical writing and experimentation with Rust, Bevy, and WebAssembly.
- **Dippy6502** — AI-assisted MOS6502 emulator with full pinout voltage visualization.
- **PanelPanic** — Rust/Macroquad Panel de Pon adaptation with WebRTC online multiplayer.
- **StickerMoji** — Web app for ordering AI-generated stickers.
- **TimursChess** — Rust/Macroquad implementation of Tamerlane Chess.